

Tutorial 6 – Decision Making and Looping

1. Using looping re-write the C++ program to check whether an alphabet given as input is vowel or, consonant. Set the loop counter to 5.

Sample output:

```
Enter an alphabet: a
a is a vowel.
Enter an alphabet: B
B is a consonant.
Enter an alphabet: E
E is a vowel.
Enter an alphabet: g
g is a consonant.
Enter an alphabet: k
k is a consonant.
```

2. Write a C++ program using a **while** loop to write out all the numbers between 100 to 500 that are divisible by 3 and 5.

Sample output:

```
105 is divisible
120 is divisible
135 is divisible
150 is divisible
165 is divisible
180 is divisible
195 is divisible
210 is divisible
225 is divisible
240 is divisible
255 is divisible
270 is divisible
285 is divisible
300 is divisible
315 is divisible
330 is divisible
345 is divisible
360 is divisible
375 is divisible
390 is divisible
405 is divisible
420 is divisible
435 is divisible
450 is divisible
465 is divisible
480 is divisible
495 is divisible

...Program finished with exit code 0
Press ENTER to exit console.
```

3. C++ program to add numbers until user enters zero.

sum of the numbers is: 6

Sample output 1:

Enter number to add (enter 0 to exit): 1
Enter number to add (enter 0 to exit): 2
Enter number to add (enter 0 to exit): 3
Enter number to add (enter 0 to exit): 0

Sample output 2:

Enter numbers to add (enter 0 to exit): 0
exit program

4. Write a program in C++ to find factorial of a given positive integer. (factorial of an integer is the product of all positive integers less than or equal to a given positive integer)

Sample Output:

Enter a Number: 6
Factorial of 6 is 720

5. Re-Write the calculator created in tutorial 5 to a Menu-Driven program to compute values until user enters 'Q'.

Sample output:

```
*****
Enter + for addition
Enter - for subtraction
Enter * for multiplication
Enter / for division
Enter Q to exit
*****
Enter an operator: (+, -, *, /, Q): *
Enter two numbers:
5
6
Product of 5 and 6 is 30
Enter an operator: (+, -, *, /, Q): -
Enter two numbers:
7
6
Difference of 7 and 6 is 1
Enter an operator: (+, -, *, /, Q): Q
end of program
```

6. Write a C++ program for PIN verification. Allow users three attempts.
assuming PIN is 1234

Sample output:

```
enter PIN: 111
enter PIN: 1234
loading...
```

Sample output:

```
enter PIN: 111
enter PIN: 222
enter PIN: 333
Blocked...
```

7. Write a program in C++ to find the first and last digit of a number.

Sample output:

```
Enter a number greater than 100: 125
The first digit of entered number is: 1
The last digit of entered number is: 5
```

(logic:

- The modulus operator % gives the remainder when divided by 10
This remainder is always the last digit
- Keep dividing the number by 10 until it becomes a single digit:
The remaining value is the first digit

)

8. Write a program in C++ to calculate product of digits of any number.

Sample output:

```
Enter a number: 123
The product of all the digits is 6
```

(logic:

% 10 gives the last digit of the number
Keep a variable product initialized to 1

Multiply each digit with variable

/ 10 This removes the last digit of the number

Negative numbers → Convert to positive using abs()
)

9. Write a C++ program to print number from 1 to 100. Numbers divisible by 3 will print "fizz", numbers divisible by 5 will print "buzz" and numbers divisible by 15 will print "fizzbuzz".

output:

1, 2, Fizz, 4, Buzz, Fizz, 7, 8, Fizz, Buzz, 11, Fizz, 13, 14, Fizz Buzz, 16, 17, Fizz, 19, Buzz, Fizz, 22, 23, Fizz, Buzz, 26, Fizz, 28, 29, Fizz Buzz, 31, 32, Fizz, 34, Buzz, Fizz, 37, 38, Fizz, Buzz, 41, Fizz, 43, 44, Fizz Buzz, 46, 47, Fizz, 49, Buzz, Fizz, 52, 53, Fizz, Buzz, 56, Fizz, 58, 59, Fizz Buzz, 61, 62, Fizz, 64, Buzz, Fizz, 67, 68, Fizz, Buzz, 71, Fizz, 73, 74, Fizz Buzz, 76, 77, Fizz, 79, Buzz, Fizz, 82, 83, Fizz, Buzz, 86, Fizz, 88, 89, Fizz Buzz, 91, 92, Fizz, 94, Buzz, Fizz, 97, 98, Fizz, Buzz,

10. Write a program in C++ to display the pattern like right angle triangle with number.

Sample output:

Input number of rows: 5

1
12
123
1234
12345

(Nested loop,

(1) outer loop keeps incrementing till number of rows is met,

(2) inner loop print numbers starting from 1 based on row number defined by outer loop)